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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

October/November 2021

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 A particle is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane. The particle moves freely under gravity.

- (a) Write down the horizontal and vertical components of the velocity of the particle at time T after projection. [2]

$$V_y = u \sin \alpha - gT$$

$$V_x = u \cos \alpha$$

At time T after projection, the direction of motion of the particle is perpendicular to the direction of projection.

- (b) Express T in terms of u , g and α . [2]

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha$$

$$\frac{dy}{dx} = \tan \alpha - \frac{gx}{u^2} \sec^2 \alpha$$

@ time T , $x = uT \cos \alpha$

$$\rightarrow \frac{dy}{dx} = \tan \alpha - \frac{g u T \cos \alpha}{u^2 \cos^2 \alpha}$$

$$= \tan \alpha - \frac{gT}{u \cos \alpha}$$

Initially, $\frac{dy}{dx} = \tan \alpha$
 $\hookrightarrow$ so at T , $\frac{dy}{dx} = \frac{-1}{\tan \alpha}$

$$\frac{-1}{\tan \alpha} = \tan \alpha - \frac{gT}{u \cos \alpha}$$

$$\frac{gT}{u \cos \alpha} = \frac{\tan^2 \alpha + 1}{\tan \alpha}$$

$$T = \left(\frac{1}{\cos^2 \alpha} \right) \left(\frac{\cos \alpha}{\sin \alpha} \right) \left(\frac{u \cos \alpha}{g} \right)$$

$$= \frac{u}{g \sin \alpha}$$

- (c) Deduce that $T > \frac{u}{g}$. [1]

$$\frac{u}{g \sin \alpha} > \frac{u}{g}$$

$$1 > \sin \alpha$$

$$\hookrightarrow \alpha \neq 90^\circ, \text{ so this is true}$$

- 2 A light spring AB has natural length a and modulus of elasticity $5mg$. The end A of the spring is attached to a fixed point on a smooth horizontal surface. A particle P of mass m is attached to the end B of the spring. The spring and particle P are at rest on the surface.

Another particle Q of mass km is moving with speed $\sqrt{4ga}$ along the horizontal surface towards P in the direction BA . The particles P and Q collide directly and coalesce. In the subsequent motion the greatest amount by which the spring is compressed is $\frac{1}{5}a$.

Find the value of k .

[6]

Diagram: A horizontal line represents a spring of natural length a and modulus of elasticity $5mg$. A particle P of mass m is attached to the end B . A particle Q of mass km is moving towards P with speed $\sqrt{4ga}$. The particles collide and coalesce. The coefficient of restitution $e = 0$.

Method: Conservation of energy

Initially:

$$KE = \frac{1}{2}(km+m)(v)^2 = \frac{1}{2}(k+1)m\left(\frac{\sqrt{4ga}k}{k+1}\right)^2 = \frac{1}{2}m\left(\frac{4gak^2}{(k+1)^2}\right)$$

Finally:

$$KE = \frac{1}{2}(km+m)(v)^2$$

$$EPE = \frac{\lambda x^2}{2\lambda} = \frac{(5mg)(\frac{a}{5})^2}{2a} = \frac{5mga^2}{50a} = \frac{1}{10}mga$$

But: At the point of maximum compression

↳ particle is turning

↳ $\therefore v=0, KE=0$

So, by CoE:

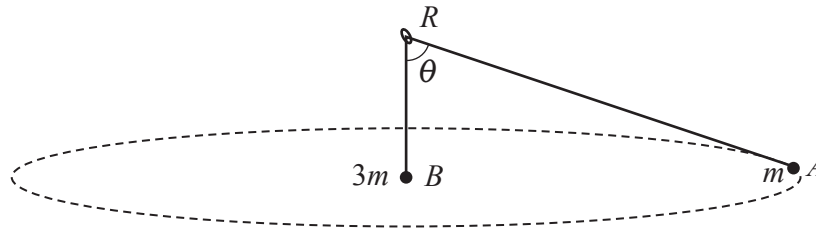
$$\frac{mga}{10} = \frac{1}{2}m\left(\frac{4gak^2}{(k+1)^2}\right)$$

$$\frac{1}{20} = \frac{k^2}{k+1} \Rightarrow 20k^2 - k - 1 = 0$$

$$(4k-1)(5k+1) = 0$$

↳ $k = \frac{1}{4}$ (as $k > 0$)

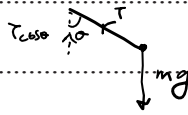
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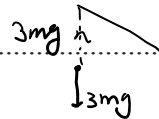
Particles A and B , of masses m and $3m$ respectively, are connected by a light inextensible string of length a that passes through a fixed smooth ring R . Particle B hangs in equilibrium vertically below the ring. Particle A moves in horizontal circles with speed v . Particles A and B are at the same horizontal level. The angle between AR and BR is θ (see diagram).

- (a) Show that $\cos \theta = \frac{1}{3}$. [2]

$$T \cos \theta = mg$$

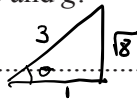


$$T = 3mg$$



$$\cos \theta = \frac{1}{3}$$

- (b) Find an expression for v in terms of a and g . [4]



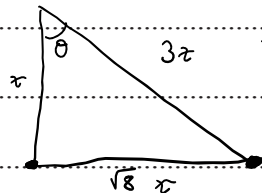
$$\frac{mv^2}{r} = T \sin \theta$$

$$\frac{mv^2}{r} = 3mg \left(\frac{2\sqrt{2}}{3} \right)$$

$$v^2 = 2\sqrt{2} rg$$

$$v^2 = 2\sqrt{2} \left(\frac{a}{\sqrt{10}} \right) g$$

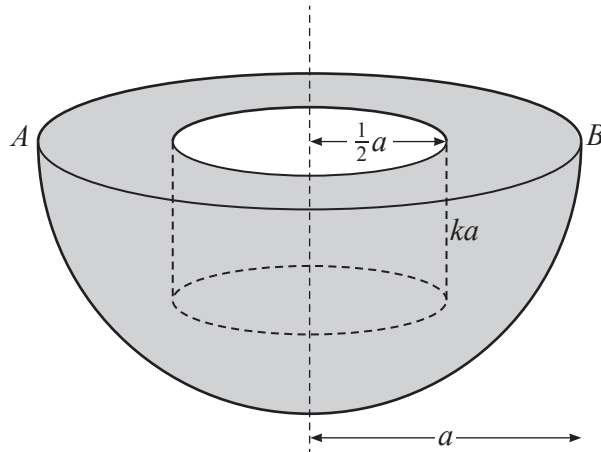
$$v = \sqrt{2ga}$$



$$x + 3x = a$$

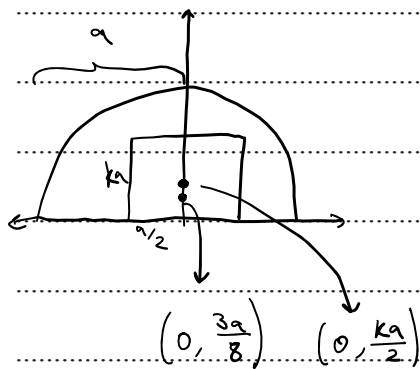
$$\hookrightarrow x = \frac{a}{4}$$

$$r = \sqrt{8}x = \frac{\sqrt{2}a}{4} = \frac{a}{2\sqrt{2}}$$



An object is formed by removing a solid cylinder, of height ka and radius $\frac{1}{2}a$, from a uniform solid hemisphere of radius a . The axes of symmetry of the hemisphere and the cylinder coincide and one circular face of the cylinder coincides with the plane face of the hemisphere. AB is a diameter of the circular face of the hemisphere (see diagram).

- (a) Show that the distance of the centre of mass of the object from AB is $\frac{3a(2-k^2)}{2(8-3k)}$. [4]



Taking moments:

$$\bar{y} \left(\frac{2}{3} \pi a^3 - \pi \left(\frac{a}{2} \right)^2 (ka) \right) (p) \\ = \left(\frac{2}{3} \pi a^3 \right) (p) \left(\frac{3a}{8} \right) + \left(-\pi \left(\frac{a}{2} \right)^2 (ka) \right) (p) \left(\frac{ka}{2} \right)$$

$$\bar{y} \left(\pi a^3 p \right) \left(\frac{2}{3} - \frac{k}{4} \right) \\ = \pi a^3 p \left(\frac{a}{4} \right) - \pi a^3 p \left(\frac{k^2 a}{8} \right)$$

$$\bar{y} \left(\frac{8-3k}{3} \right) = \frac{2a-k^2 a}{2}$$

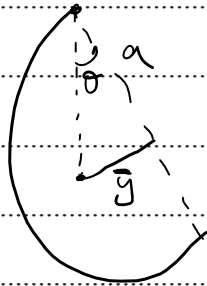
$$\bar{y} = \frac{6a - 3k^2 a}{16 - 6k}$$

$$= \frac{3a(2-k^2)}{2(8-3k)}$$

When the object is freely suspended from the point A , the line AB makes an angle θ with the downward vertical, where $\tan \theta = \frac{7}{18}$.

(b) Find the possible values of k .

[3]



$$\tan \theta = \frac{7}{18}$$

$$\frac{7}{18} = \frac{y}{a}$$

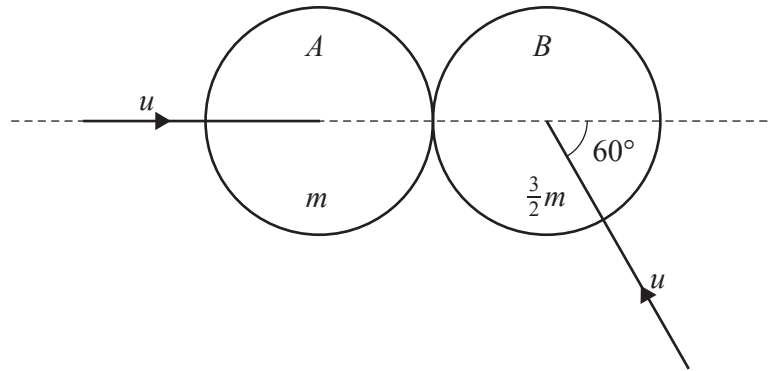
$$\left(\frac{3}{2}\right)\left(\frac{2-k^2}{8-3k}\right) = \frac{7}{18}$$

$$108 - 54k^2 = 112 - 42k$$

$$0 = 54k^2 - 42k + 4$$

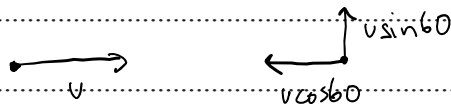
$$= (3k-2)(9k-1)(2)$$

$$\rightarrow k = \frac{2}{3} \text{ or } \frac{1}{9}$$



Two uniform smooth spheres A and B of equal radii have masses m and $\frac{3}{2}m$ respectively. The two spheres are each moving with speed u on a horizontal surface when they collide. Immediately before the collision A 's direction of motion is along the line of centres, and B 's direction of motion makes an angle of 60° with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{2}{3}$.

- (a) Find the angle through which the direction of motion of B is deflected by the collision. [6]



Only horizontal components are affected by the collision

$$\text{CoM: } mu = \left(\frac{3m}{2}\right)\left(\frac{u}{2}\right) = mv_1 + \frac{3m}{2}v_2$$

$$\frac{u}{4} = v_1 + \frac{3}{2}v_2$$

$$\text{Newtons: } \frac{v_2 - v_1}{\frac{3u}{2}} = \frac{2}{3}$$

$$v_2 - v_1 = u$$

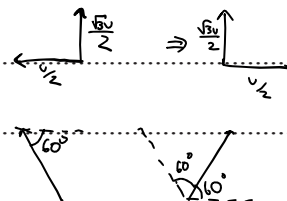
$$+ v_1 + \frac{3}{2}v_2 = \frac{u}{4}$$

$$\frac{5}{2}v_2 = \frac{5u}{4}$$

$$\boxed{v_2 = \frac{u}{2}}$$

$$v_1 = -\frac{u}{2}$$

Considering B post collision



It is deflected through

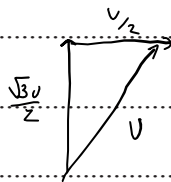
a $\boxed{60^\circ}$ angle

- (b) Find the loss in the total kinetic energy of the system as a result of the collision. [3]

Initially:

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{3m}{2}\right)(v^2) = \frac{5}{4}mv^2 \text{ J}$$

Find:



$$\rightarrow B \text{ has } \frac{1}{2}\left(\frac{3m}{2}\right)v^2 = \frac{3m}{4}v^2 \text{ J}$$

$$\leftarrow \frac{v}{2} \rightarrow A \text{ has } \frac{1}{2}(m)\left(\frac{v}{2}\right)^2 = \frac{mv^2}{8} \text{ J}$$

$$Loss = \frac{5mv^2}{4} - \frac{3mv^2}{4} - \frac{mv^2}{8} = \boxed{\frac{3mv^2}{8}} \text{ J}$$

- 6 A particle P of mass 2 kg moves along a horizontal straight line. The point O is a fixed point on this line. At time $t\text{ s}$ the velocity of P is $v\text{ ms}^{-1}$ and the displacement of P from O is $x\text{ m}$.

A force of magnitude $\left(8x - \frac{128}{x^3}\right)\text{ N}$ acts on P in the direction OP . When $t = 0$, $x = 8$ and $v = -15$.

- (a) Show that $v = -\frac{2}{x}(x^2 - 4)$. [5]

$$a = \left(8x - \frac{128}{x^3}\right)\left(\frac{1}{x}\right) = 4x - \frac{64}{x^3}$$

By Chain rule:

$$\frac{\left(\frac{dv}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{4x - \frac{64}{x^3}}{dx/dt}$$

$$v dv = 4x - \frac{64}{x^3} dx$$

$$\int v dv = \int 4x - \frac{64}{x^3} dx$$

$$\frac{v^2}{2} = 2x^2 + \frac{32}{x^2} + C$$

$$v^2 = 4x^2 + \frac{64}{x^2} + C$$

$$\text{sub } (8, -15)$$

$$225 = 4(64) + 1 + C$$

$$C = -32$$

$$v^2 = 4x^2 + \frac{64}{x^2} - 32$$

$$= (2x)^2 + \left(\frac{8}{x}\right)^2 - 2(2x)\left(\frac{8}{x}\right)$$

$$v^2 = \left(2x - \frac{8}{x}\right)^2$$

$$v^2 = \left(\frac{2}{x}(x^2 - 4)\right)^2$$

$$v = \pm \left(\frac{2}{x}(x^2 - 4)\right)$$

$\Rightarrow$ we know that when $x = 8$, $v = -15$

$\hookrightarrow$ we take the negative root

$$v = -\frac{2}{x}(x^2 - 4)$$

(b) Find an expression for x in terms of t .

[4]

$$\frac{dx}{dt} = \frac{-2}{x} (x^2 - 4)$$

$$\frac{-x}{2} \left(\frac{1}{x^2 - 4} \right) dx = dt$$

$$\int \frac{x}{x^2 - 4} dx = \int -2 dt$$

$$\int \frac{x}{x^2 - 4} dx = -2t + C$$

$$\begin{aligned} \hookrightarrow \text{let } u &= x^2 - 4 \\ du &= 2x dx \\ \hookrightarrow dx &= \frac{du}{2x} \end{aligned}$$

$$\begin{aligned} \int \frac{x}{x^2 - 4} dx &= \int \frac{x}{u} \left(\frac{du}{2x} \right) \\ &= \frac{1}{2} \int \frac{1}{u} du \\ &= \frac{1}{2} \ln|u| + C \end{aligned}$$

$$= \frac{1}{2} \ln|x^2 - 4| + C$$

$$\therefore \frac{1}{2} \ln|x^2 - 4| = -2t + C$$

$$\textcircled{a} t = 0, x = 8$$

$$\frac{1}{2} \ln|60| = C$$

$$\frac{1}{2} \ln|x^2 - 4| = -2t + \frac{1}{2} \ln 60$$

$$\Rightarrow 64 - 4 = 60e^{-4t}, \text{ not } -60e^{-4t}$$

$$\ln|x^2 - 4| = -4t + \ln 60$$

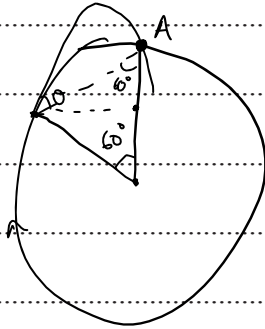
$$|x^2 - 4| = e^{-4t + \ln 60}$$

$$|x^2 - 4| = 60e^{-4t}$$

$$\therefore x^2 - 4 = 60e^{-4t} \Rightarrow x = \sqrt{4 + 60e^{-4t}} \quad \text{as}$$

- 7 One end of a light inextensible string of length a is attached to a fixed point O . The other end of the string is attached to a particle P of mass m . The particle P is held vertically below O with the string taut and then projected horizontally. When the string makes an angle of 60° with the upward vertical, P becomes detached from the string. In its subsequent motion, P passes through the point A which is a distance a vertically above O .

- (a) The speed of P when it becomes detached from the string is V . Use the equation of the trajectory of a projectile to find V in terms of a and g . [4]



$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

$$\text{initially: } x = y = 0 \Rightarrow (0, 0)$$

$$\text{At A: } x = a \sin 60 = \frac{\sqrt{3}}{2}a \Rightarrow \left(\frac{\sqrt{3}a}{2}, \frac{a}{2} \right)$$

$$y = a - a \cos 60 = \frac{a}{2}$$

a point

$$\rightarrow \text{Also } \theta = 60^\circ$$

$$\rightarrow \text{Substituting } \left(\frac{\sqrt{3}a}{2}, \frac{a}{2} \right) \text{ and } \theta = 60^\circ:$$

$$\frac{a}{2} = \frac{\sqrt{3}}{2}a \tan 60 - \frac{g \left(\frac{\sqrt{3}a}{2} \right)^2}{2(v^2)} \sec^2(60)$$

$$a = \sqrt{3} \left(\frac{\sqrt{3}}{2}a \right) - \frac{g \left(\frac{3a^2}{4} \right)}{v^2} \quad (4)$$

$$a = 3a - \frac{3a^2g}{v^2}$$

$$\frac{3a^2g}{v^2} = 2a$$

$$2v^2 = 3ga$$

$$v = \sqrt{\frac{3}{2}ga}$$

- (b) Find, in terms of m and g , the tension in the string immediately after P is initially projected horizontally. [4]

$$\frac{mv^2}{r} = \frac{mv^2}{r} - 2mg + 2mg \cos \theta$$

$$\hookrightarrow @ \theta = 120^\circ, v = \sqrt{\frac{3}{2}ga}$$

$$\frac{m(\frac{3}{2}ga)}{a} = \frac{mv^2}{a} - 2mg + 2mg \cos 120$$

$$\frac{3}{2}g = \frac{v^2}{a} - 3g$$

$$v = \sqrt{4.5ga}$$

$$T = \frac{mv^2}{r} - 2mg + 3mg \cos \theta$$

$$\hookrightarrow \theta = 0, v = \sqrt{4.5ga}$$

$$T = \frac{m(4.5ga)}{a} - 2mg + 3mg$$

$$= 4.5mg - 2mg + 3mg$$

$$\boxed{\approx 5.5mg}$$

[illegible]

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